

THE SECRET MYSTERY BOXES: WELCOME TO ALGEBRA!

Crack Hidden Codes and Solve for Unknowns

Grade 4 Introduction to Algebra Guide

Did you know you can play detective right inside math class?

*Let's discover how to find secret numbers hiding behind shapes
and letters!*

You are Now a Math Detective!

Have you ever opened a wrapped birthday present and tried to guess what was inside?

In math up until today, you have usually seen all the numbers clearly. For example, your teacher writes down $5 + 3 = 8$, and everything is completely out in the open.

But today, we are going to start a brand new adventure called **Algebra**! In Algebra, numbers like to hide in secret boxes, behind blank question marks, or underneath funny colorful shapes.

Your job as a math detective is to read the clues left behind in the equation and figure out exactly what number is hiding from us!

What is an "Unknown"?

When a number is hiding from us, mathematicians give it a very cool name. They call it an **Unknown**!

An unknown is simply a blank placeholder for a number we don't know yet, but we can easily figure out using our logic brains.

Look at this riddle statement below:

$$[\textit{Mystery Box}] + 4 = 10$$

Think about it like a balance scale. What number must you put inside that empty brown mystery box so that when you add 4 to it, it perfectly hits 10?

The Answer:

It must be 6! Because $6 + 4 = 10$. You just solved your very first unknown algebra problem!

Trading Boxes for Letters

Drawing mystery boxes or stars takes a long time, right? Let's use a shortcut!

Because mathematicians can be a little bit lazy with drawing, they decided to trade mystery boxes for simple lowercase letters. They use alphabet letters like x , y , or a as the masks for our hiding numbers!

So instead of writing: $[\text{Box}] + 3 = 7$

We write it elegantly like this:

$$x + 3 = 7$$

Don't let the letter x scare you! Whenever you see an x hanging out in a math problem, just pretend it's a mystery box whispering, "Guess what number I am!"

Cracking the Addition Code

Let's practice solving for the unknown letter x using our logic steps. Let's look at this equation:

$$x + 5 = 12$$

This clue tells us: "Some secret number plus 5 equals 12." How do we unpack this?

The Backward Counting Method (Subtraction Trick):

To find out what's hiding, we can do the opposite action! Let's count backwards by subtracting 5 from 12.

$$12 - 5 = 7$$

So, our secret unknown letter must be $x = 7$!

Let's check it: Does $7 + 5 = 12$? Yes, it does! The case is solved!

The Great Balance Scale Rule

In Algebra, the equal sign ($=$) acts exactly like a playground see-saw or an old-fashioned **balance scale**.

Everything on the left-hand side of the scale must weigh the exact same amount as everything on the right-hand side of the scale to keep it perfectly balanced!

Left Side Weight		Right Side Weight
$y + 2$	$=$ MUST MATCH	9

If the right side says 9, the left side *must* total up to 9. What number must y be to make that happen? y has to be 7, because $7 + 2 = 9$.

Golden Rule: Whatever number you choose for the unknown letter must keep both sides of the equal sign perfectly even and happy!

Solving Subtraction Mysteries

What happens when the unknown number faces a subtraction minus sign? Let's check!

Let's look at this code sentence:

$$x - 3 = 5$$

This translates to: "A hidden total number of apples minus 3 apples leaves us with 5 apples."

To figure out our giant starting number of apples before any were eaten, we can bundle the remaining apples back together using addition:

$$5 + 3 = 8$$

So, our unknown value is $x = 8$! Because starting with 8 and taking away 3 lands us exactly on 5.

Real World Story: The Candy Jar Mystery

Let's use our new algebraic detective skills to help a boy named Rohan solve a candy problem at home!

Rohan has an opaque jar filled with a secret number of strawberry candies (c). His friend Maya gives him 6 more candies. Rohan counts all his candies together and finds out he now has 15 candies in total!

How many candies were originally hiding inside Rohan's jar?

- **Step 1: Write down the algebra code:** $c + 6 = 15$
- **Step 2: Do the inverse opposite action:** Subtract 6 from 15.
 $15 - 6 = 9$
- **Step 3: State the final unknown answer:** $c = 9$ candies!

Fantastic work! Rohan originally had exactly 9 sweet candies hiding inside his jar!

Workbook Practice: Try it Out!

Grab your favorite pencil and let's work through these problems together!

Problem 1: Solve for the unknown letter "a"

$$a + 4 = 11$$

- Think: What number plus 4 equals 11? (Or do $11 - 4$)
- Your Answer: $a =$ _____

Problem 2: Solve for the unknown letter "z"

$$z - 5 = 4$$

- Think: What number minus 5 leaves you with 4? (Or do $4 + 5$)
- Your Answer: $z =$ _____

Answer Check: Problem 1 is 7! Problem 2 is 9! Did you get them right? You are amazing!

You Are an Official Algebra Expert!

You have successfully unlocked the entry gate to the wonderful world of higher mathematics! Let's review our official detective checklist:

- **The Unknown:** A hidden number that we are searching for in an equation.
- **Letter Masks:** Lowercase letters like x , y , and a are just friendly placeholders for mystery boxes.
- **The Scale Rule:** An equation must always stay perfectly balanced on both sides of the equal sign.
- **Opposite Actions:** Use subtraction to undo addition, and use addition to undo subtraction!

The Master Graduation Challenge!

Can you crack this ultimate secret code?

$$x + 10 = 30$$

What is the value of x ?

(Write your answer down and show it proudly to your teacher or parents to prove you are a true math detective!)

Detective Practice Zone

Become an ultimate code cracker! Solve these 12 algebra mysteries by finding the secret value of each unknown letter.

1. Solve for x: $x + 3 = 8$

Answer: x =

.....

2. Solve for a: $a + 7 = 15$

Answer: a =

.....

3. Solve for y: $y - 4 = 6$

Answer: y =

.....

4. Solve for m: $m - 2 = 9$

Answer: m =

.....

5. Solve for b: $b + 6 = 14$

Answer: b =

.....

6. Solve for k: $k - 5 = 5$

Answer: k =

.....

Detective Practice Zone (Continued)

7. Solve for p: $p + 9 = 20$

Answer: p =

8. Solve for n: $n - 8 = 3$

Answer: n =

9. Solve for w: $w + 12 = 18$

Answer: w =

10. Solve for h: $h - 10 = 7$

Answer: h =

11. Word Mystery: Leo has a secret number of marbles m . After his sister gives him 4 more, he has 13 marbles. Write the equation and solve for m .

Equation & Answer:

12. Word Mystery: A box of cupcakes had a secret total number of cupcakes c . After 5 cupcakes were eaten, there are 7 left. Write the equation and solve for c .

Equation & Answer: